Electricity Demand Forecasting and Feature Engineering

Practical ML concepts for a next-day Texas electricity demand workflow

Document purpose: This document is prepared as a searchable knowledge-base source for a SmartGrid AI RAG prototype. It is an educational synthesis, not an official ERCOT, utility, or government publication.

Forecasting objective

Electricity demand forecasting estimates future electric load using historical load, weather, calendar effects, and other relevant information. A next-day forecast should specify whether it predicts an hourly profile, daily average, daily peak, or a zone-level total.

The prediction target must match the dashboard label. For example, a daily peak-demand forecast should not be described as an hourly forecast. The unit (MW, MWh, or another measure) must also be explicit.

Potential input features

Historical demand: previous hour, previous day, previous week, rolling mean, rolling maximum, and recent trend.

Weather: temperature, humidity, wind speed, precipitation, and forecast extreme-weather indicators.

Calendar: hour of day, day of week, weekend, month, season, and public holidays where relevant.

Geography and system context: ERCOT zone, county, population or customer mix where available, and known system events.

Feature engineering

Lag features capture recent demand patterns. For hourly data, examples include demand one hour ago, 24 hours ago, and 168 hours ago. Rolling averages can smooth noise, but they must be calculated using only observations available before the forecast time.

Cooling-degree and heating-degree features summarize how far temperature is above or below a chosen comfort reference. The reference threshold should be selected and documented for the dataset and use case; it is not a universal constant.

Categorical features such as zone or day of week may require encoding. Continuous features may need scaling for some algorithms. Tree-based models often need less feature scaling than linear or distance-based models.

Model evaluation

Use chronological train, validation, and test splits for time-dependent data. Randomly shuffling rows can leak future patterns into training and produce overly optimistic results.

MAE measures average absolute prediction error in the target's units. RMSE penalizes large errors more strongly. R-squared describes the proportion of variance explained relative to a baseline, but it should not be used alone.

Compare models on the same held-out time period and report the forecast horizon, target definition, units, and baseline. Consider performance during extreme-weather periods separately because average performance can hide rare high-impact errors.

Avoiding data leakage

Every feature must be available at the time the forecast would be made. Using actual next-day weather or future demand values as inputs creates leakage. Fit preprocessing transformations on the training data only, then apply them to validation and test data.

Keep county, zone, and date definitions consistent when joining weather, demand, and outage data. Check for duplicate keys and mismatched time zones.

Key takeaways

- Define the exact demand target, horizon, and units before training.
- Use time-aware splits and features available at prediction time.
- MAE and RMSE are useful complementary regression metrics.
- Evaluate extreme-weather periods and compare against a simple baseline.

References and further reading

• ERCOT Grid Information and Load Data — https://www.ercot.com/gridinfo

• NERC Reliability Assessments — https://www.nerc.com/our-work/assessments

Prepared 9 November 2026. Verify current operational information with the linked official sources before making real-world decisions.